

Title

Evaluating a structured digital platform with nursing support for reducing dietary sodium in kidney disease: a protocol for implementation science research

Abstract

Excessive dietary sodium intake accelerates renal disease progression and hypertension in patients with chronic kidney disease. While evidence-based education can improve dietary awareness, adoption of sustainable behavioral change remains challenging. This mixed-methods implementation study will evaluate a technology-enhanced nursing intervention designed to support sodium restriction in patients with stages 3-4 chronic kidney disease. The protocol combines quantitative measures (24-hour urinary sodium as primary endpoint, blood pressure, and knowledge assessments) with qualitative interviews exploring facilitators and barriers to engagement. Approximately 100 participants will be randomized 1:1 to receive either enhanced digital coaching with nurse-led support or usual care comparator. Data collection spans 12 months. Mixed-methods analysis will integrate quantitative outcomes with qualitative findings to identify implementation factors influencing behavior change and inform future scaling.